CS 2490 – Homework 7

Turn in by 2026/10/24 at 11:59pm

You **must** turn in your assignment as a zip file entitled: *lastname_firstname_hw7.zip*. It will contain a pdf file *lastname_firstname_hw7.pdf* with your solutions to the exercises, and a JSON file *lastname_firstname_hw7_ex4.json* for the PDA of Exercise 4.

Exercise 1. In this exercise, we work over the alphabet $\Sigma = \{a, b\}$. Show that $L = \{a^n b^m \mid n > m \geq 0\}$ is a context-free language. In other words, exhibit a context-free grammar that generates L and give a quick justification.[1] *Hint: remember that you can have more than one variable symbol in your grammar.*

[1] We don't ask for a formal rigorous proof, but explain your thought process in crafting the grammar you propose.

Exercise 2. In this exercise, we work over the alphabet $\Sigma = \{a, b, c\}$.

(i) Determine the language $\mathrm{L}(G)$ generated by the grammar $G = (\Sigma, \{S\}, S, P)$ where P contains the following production rules:

$$S \to \varepsilon, \quad S \to a, \quad S \to b, \quad S \to c$$
$$S \to aSa, \quad S \to bSb, \quad S \to cSc$$

(ii) Is $\mathrm{L}(G)$ a context-free language? Justify your answer.

(iii) Choose a word of length at least 6 in the language you found and give a derivation for it from S in G. (We expect an answer in the form of a sequence $S \Rightarrow \cdots \Rightarrow w$ where w is the word you chose.)

Exercise 3. In this exercise, we work over the alphabet $\Sigma = \{a, b, c\}$. In Recitation 8 (Exercise 3), you proved that the following language is not regular:

$$L = \{a^i b^j c^k \mid i = j \text{ or } j = k\}.$$

(i) Construct a context-free grammar that generates L and give a quick justification.

(ii) Draw a derivation tree for the word *aabbbccc* in your grammar, and give the corresponding leftmost derivation.

(iii) Show that your grammar is ambiguous.[2]

[2] This is not a coincidence: every context-free grammar that generates L is ambiguous, i.e., L is *inherently ambiguous*. We don't ask you to prove that.

Exercise 4. Construct a PDA $\mathscr{A}$ that recognizes the language of the previous exercise. *(Simply draw the PDA, you don't have to justify that it indeed recognizes the correct language.)*

Include the JSON file of your PDA (produced with the web application) in your submission.